ADEMP protocol: 07-gompertz-evaluation

2026-05-07 16:55 PDT

This document is the working protocol for the exploratory simulation study that underlies `analysis/report/07-gompertz-evaluation/`. The study is exploratory, and its findings should be confirmed in an independent study. ADEMP structure follows Morris, White and Crowther (2019).

Common infrastructure

Trial design. Hendrickson hybrid OL+BDC design, 16-week total trial duration, 8-week open-label active drug + 4-week blinded discontinuation + 4-week crossover.

Sample sizes. $N \in \{35, 70, 100\}$.

Drug context. Prazosin for PTSD nightmares, calibrated to the Hendrickson 2020 reference parameters at the saturating asymptote.

Replicates. $n_{\text{reps}} = 1000$ per cell for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error cells.

Random-number control. Per-replicate seed via cell-descriptor hash; base seed in `01-base-seed.txt`.

Reporting standard. Morris-White-Crowther ADEMP with MCSE columns for every reported performance measure.

Calibration to common effect size. All four DGP families share the same marginal Cohen's-style standardised effect size at week 8 in the open-label phase, computed as the marginal on-drug Cor(B, BR) in the saturated regime. Each family's parameters are fitted to match this effect size before the production grid runs, via the calibration sub-study described in Study 0 below.

Study 0. Effect-size calibration sub-study

A. Fit the parameters of each of the four DGP families (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise- linear breakpoint) so that all four produce the same marginal on-drug Cor(B, BR) at week 8.

D. Single-component BR-only generative model under each of the four families, with population-level participant heterogeneity but no PB or TV component (those are added in Study 1).

E. The set of family-specific parameters that produces matched marginal effect size.

M. Numerical optimisation via `optim`, with an analytic Jacobian where available.

P. Convergence to the calibration target within $|d| < 0.01$ on Cohen's-style effect size; reproducibility across random seeds.

Study 1. Power, bias, and type I error across families

A. Quantify how the standard linear-mixed analysis's inferential properties (power, type I error, bias) depend on the DGP family used to generate the data.

D. Four DGP families on the BR factor only:

1. *Modified Gompertz* (the pmsimstats default): $BR(t) = m \exp(-d \exp(-rt))$ with vertical-offset adjustment.
2. *Symmetric logistic*: $BR(t) = m/(1 + \exp(-r(t - t_0)))$.
3. *Hyperbolic-tangent*: $BR(t) = (m/2)(1 + \tanh(r(t - t_0)))$.
4. *Piecewise-linear breakpoint*: $BR(t) = m \cdot \min(1, t/t_0)$.

PB and TV components are held at their Hendrickson reference specifications (modGompertz with calibrated parameters), so the family axis varies BR alone. Each family is calibrated to the matched marginal effect size from Study 0.

E. Primary estimand: $\hat{\beta}_{bm:D}$ from the linear- mixed analysis. Secondary estimand: posterior class entropy is not applicable here (single-component DGPs), but the empirical SD of $\hat{\beta}_{bm:D}$ across replicates is reported as a finite-sample efficiency measure.

M. Single analysis: linear mixed-effects model `Sx ~ bm + t + Dbc + bm:Dbc + (1 | ptID)` with `corCAR1(form = ~t|ptID)`.

P. Power for $H_0 : \beta_{bm:D} = 0$, type I error under $\beta_{bm:D} = 0$, bias of $\hat{\beta}_{bm:D}$, empirical SE across replicates, model SE, nominal-95% CI coverage, convergence rate. MCSE attached to every estimate.

Working prediction. Power varies less than $\pm 5\%$ across the four families at fixed sample size and effect size, provided each is calibrated to the matched marginal effect size. Type I error remains within $\pm 1\%$ of nominal across families. Bias is small at all four families when the analysis model's continuous-decay carryover specification matches the DGP's saturating-shape regime.

Total cells. 4 families $\times$ $3N$ $\times$ 3 effect sizes $\times$ $4t_{1/2,\mathrm{DGP}} = 144$ cells.

Study 2. Sloppiness and parameter identifiability

A. Characterise the parameter-redundancy structure (sloppiness) of the modified Gompertz form fitted to short N-of-1 trajectories.

D. Single Gompertz BR generative model at the Hendrickson reference; eigenvalue analysis of the parameter Hessian at the maximum-likelihood point, computed analytically.

E. The set of eigenvalues of the parameter Hessian, ordered by magnitude. Following @jagadeesan2023sloppiness, the sloppiness is characterised by the ratio of the largest to the smallest eigenvalue.

M. Numerical Hessian computation at the MLE for each replicate; analytical comparison to the unified-Richards reparameterisation of @tjorve2010richards.

P. Mean and variance of each eigenvalue across replicates; the sloppiness ratio; the sample-size scaling of the smallest eigenvalue.

Deliverables

- Driver scripts at `analysis/scripts/gompertz-evaluation/`.
- ADEMP results tables per study with MCSE columns.
- Figures: power/bias/type-I plots faceted by DGP family; sloppiness eigenvalue spectrum.

Protocol change log

`02-deviations-log.md` records changes to the cells, estimands, methods, or performance measures specified in this working protocol.

Source: `~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/analysis/scripts/gompertz-evaluation/00-adem`